21 $v^2 = u^2 + 2as$

$u = 12 \quad a = -3 \quad s = 18$

(a) Work out a value of v.

$v^2 = 12^2 + 2 \times (-3) \times 18$

$v^2 = 144 - 108 = 36$

$v = 6$ (also -6 is a possible value).

1 mark
Give a square root of 36

1 mark
Substitute with the negative sign

6

(2)

(b) Make s the subject of $v^2 = u^2 + 2as$

$v^2 = u^2 + 2as$

$-u^2 \qquad -u^2$

$v^2 - u^2 = 2as$

$\div 2a \qquad \div 2a$

$$\frac{v^2 - u^2}{2a} = s$$

1 mark
Subtract u^2 from both sides

1 mark
Divide both sides by 2a

(2)

(Total for Question 21 is 4 marks)